

Labwork 2

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Advanced HPC – USTH
September 30, 2026

1 Device Name

`cuda.detect()` finds the supported GPU; `cuda.select_device(0)` selects it. Device is: NVIDIA GeForce RTX 3050 Laptop GPU.

```
cuda.detect()

Found 1 CUDA devices
id 0      b'NVIDIA GeForce RTX 3050 Laptop GPU'      [SUPPORTED]
        Compute Capability: 8.6
        PCI Device ID: 0
        PCI Bus ID: 1
        UUID: GPU-5fc600a1-4f79-5fba-277f-8f295050e399
        Watchdog: Enabled
        Compute Mode: WDDM
        FP32/FP64 Performance Ratio: 64
Summary:
    1/1 devices are supported

True
```

Figure 1: `cuda.detect()` showing the CUDA device.

```
gpu = cuda.select_device(0)
gpu.id, gpu.name

(0, b'NVIDIA GeForce RTX 3050 Laptop GPU')
```

Figure 2: `cuda.select_device(0)` returning device id and name.

2 Core Info

Compute capability 8.6 (Ampere), 16 multiprocessors.

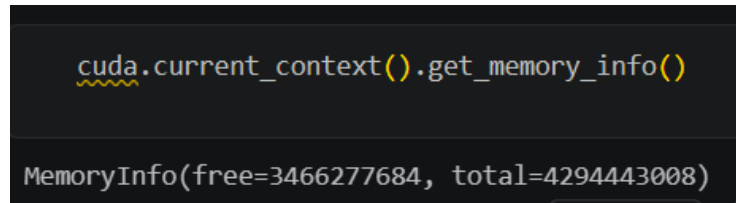
```
gpu.compute_capability, gpu.MULTIPROCESSOR_COUNT

((8, 6), 16)
```

Figure 3: Compute capability and multiprocessor count from device.

3 Memory Info

`get_memory_info()` reports free and total device memory in bytes.



```
cuda.current_context().get_memory_info()  
  
MemoryInfo(free=3466277684, total=4294443008)
```

Figure 4: Free and total memory returned by `get_memory_info()`.